

I. Convolution - Examples.

Recall : $x(t)$ and $h(t)$: two given signals.

$$y = x * h$$

$$y(t) = \int_{\tau=-\infty}^{+\infty} x(\tau) \cdot h(t-\tau) d\tau$$

$$= \int_{\tau=-\infty}^{+\infty} h(\tau) \cdot x(t-\tau) d\tau$$

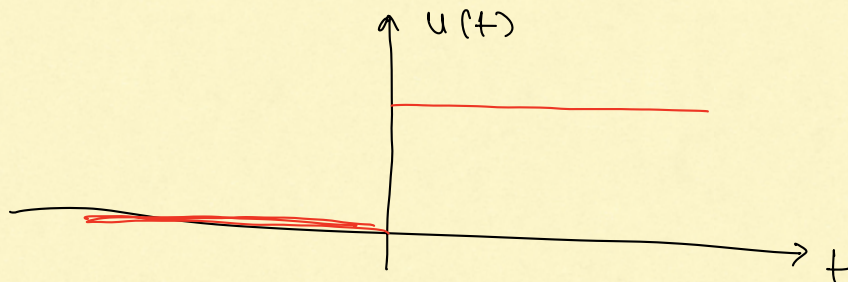
This is because convolution is commutative:

$$x * h = h * x.$$

Notation: We denote the unit step function or Heaviside step function as $u(t)$.

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Named after Oliver Heaviside
(mathematician & electrical engineer)

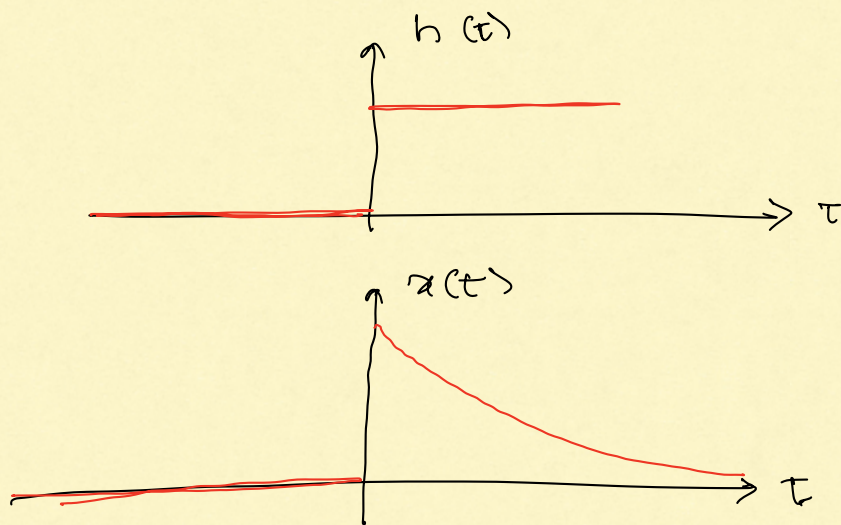
$$u(t) = \begin{cases} 1, & t > 0 \\ 0, & t < 0 \\ 1 \text{ or } 1/2 & \text{at } t=0. \end{cases}$$



Ex. 1 (from GWN Chap. 2, Ex. 2.8)

$$h(t) = u(t)$$

$$x(t) = e^{-at} u(t), \quad a > 0 \text{ is some constant.}$$



Use the flip and drag method on h for convolution.

It is straight-forward to see that

$$\text{for } t < 0: \quad x(t) \cdot h(t-\tau) = 0 \quad \forall \tau$$

$$\text{for } t > 0: \quad x(t) \cdot h(t-\tau) \neq 0 \text{ only for } 0 \leq \tau \leq t.$$

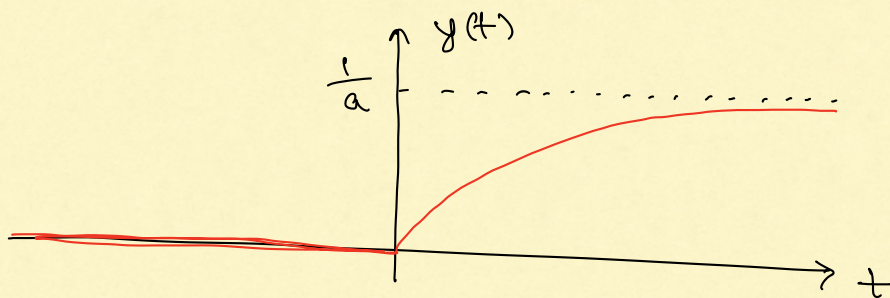
$$\therefore \text{ for } t < 0: \quad y(t) = 0$$

$$\text{for } t > 0: \quad y(t) = \int_{\tau=0}^t x(\tau) d\tau = \int_0^t e^{-a\tau} d\tau$$

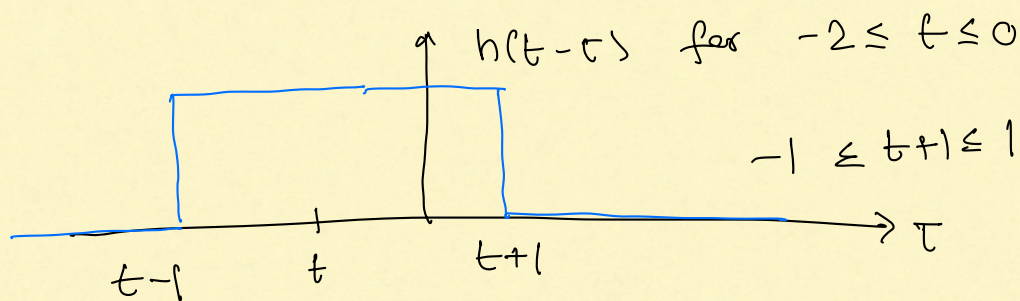
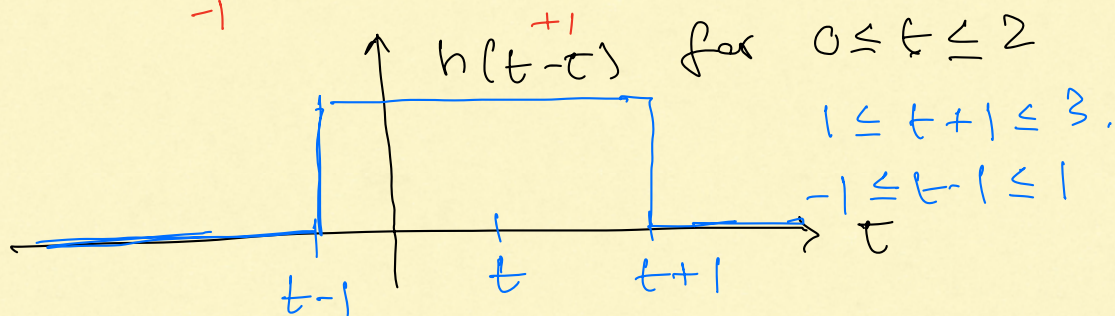
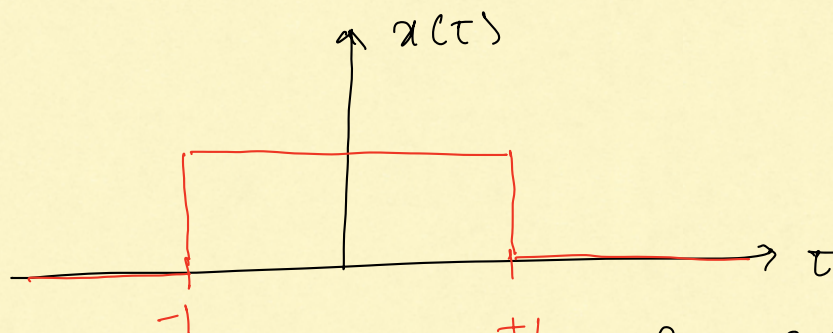
$$= \left[\frac{e^{-a\tau}}{-a} \right]_0^t$$

$$= \frac{1}{a} (1 - e^{-at}).$$

$$\Rightarrow y(t) = \frac{1}{a} (1 - e^{-at}) \cdot u(t).$$

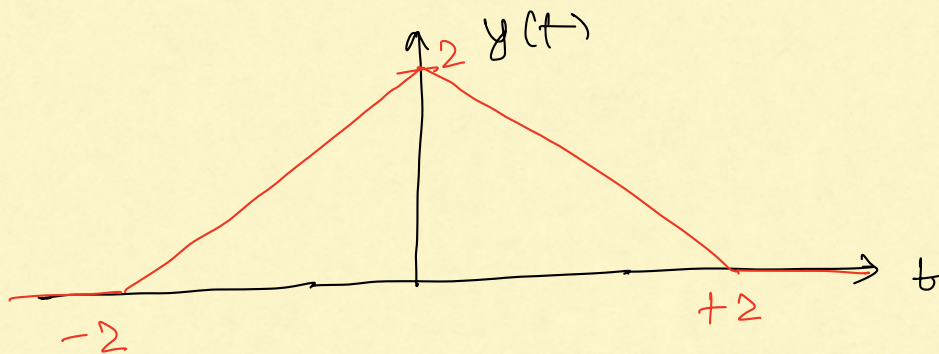


Ex 2: $x(t) = h(t) = \begin{cases} 1 & \text{if } -1 \leq t \leq +1 \\ 0 & \text{otherwise} \end{cases}$



From discussion:

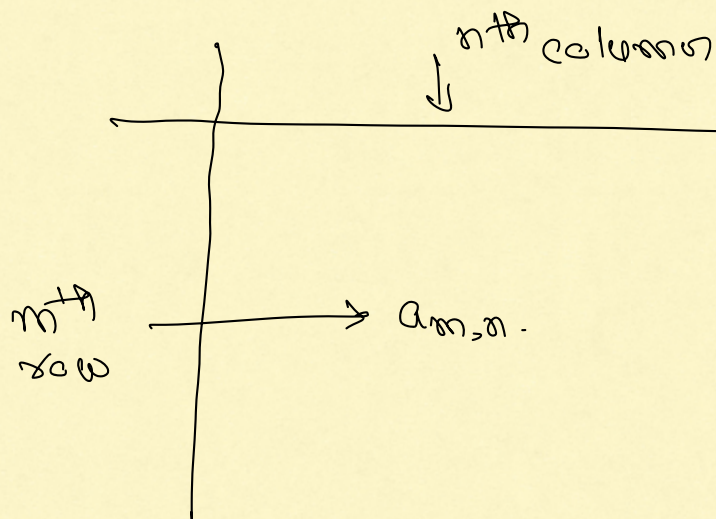
$$y(t) = \begin{cases} 0 & \text{if } t \leq -2 \text{ or } t \geq +2 \\ 2+t & \text{if } -2 \leq t \leq 0 \\ 2-t & \text{if } 0 \leq t \leq 2. \end{cases}$$



IP. About Interchanging the order of double summation and double integration.

(a) Double Summation

Suppose $a_{m,n}$ is a doubly indexed sequence of numbers with $m \geq 1$ and $n \geq 1$.



Q: Under what conditions is $\sum_{m=1}^{\infty} \sum_{n=1}^{\infty} a_{m,n}$ well defined?

Ex:

	1	2	3	4	5	...
1	1	-1	0	0	0	0
2	0	1	-1	0	0	0
3	0	0	1	-1	0	0
4	0	0	0	1	-1	0
...	...	...	...	...	...	...

Körner "Fourier Analysis"
Chap 48.

Ex:

	1	2	3	4	5	...
1	1	$-\frac{1}{2}$	$-\frac{1}{4}$	$-\frac{1}{8}$	$-\frac{1}{16}$	...
2	0	$\frac{1}{2}$	$-\frac{1}{4}$	$-\frac{1}{8}$	$-\frac{1}{16}$	...
3	0	0	$\frac{1}{4}$	$-\frac{1}{8}$	$-\frac{1}{16}$	...
4	0	0	$\frac{1}{4}$	$-\frac{1}{8}$	$-\frac{1}{16}$	...
5	0	0	$\frac{1}{4}$	$-\frac{1}{8}$	$-\frac{1}{16}$	...
...	...	...	...	...	...	...

$$\text{If } \sum_n \left(\sum_m |a_{m,n}| \right) < \infty \quad \text{or} \quad \sum_m \left(\sum_n |a_{m,n}| \right) < \infty$$

$$\text{then: } \sum_n \left(\sum_m a_{m,n} \right) = \sum_m \left(\sum_n a_{m,n} \right)$$

and the double sum is well defined.

(b) About double integration (iterated integrals).

For Riemann integration:

There is no simple criterion to determine

when

$$\int_x \left(\int_y f(x,y) dy \right) dx = \int_y \left(\int_x f(x,y) dx \right) dy.$$

For more information: See Chap 48 of Körner.

In this course: It is okay to change the order of integration.

Syllabus for Quizzes: Lectures 11-16.

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(Monday 9am)